STONE

Professional Credible and Successful
Chief Command Set

“Hand Shake” Command
“0x00” Command

Function: Please send “Hand Shake” command to the terminal by software before you further operate the software with terminal, so the terminal starts to communicate with your equipment.
Note: After receiving the feedback from the terminal, the connection is successful.

Command: “0xAA+0x00+ 0xCC 0x33 0xC3 0x3C”
Feedback: “AA 00 ‘OK_V*.*’ P1 P2 P3 Pic_ID CC 33 C3 3C”

Explain:

	Introduction	Example	Note
AA	Frame head		Every command must have the frame head and the frame end
‘OK_V*.*’	Terminal Version	0x04 0x2E 0x04	Terminal Version is V4.4
P1	Resolution	0x01	For example: 3.5 inch
P2	Baud rate code	0x07	Mean 115200 bps
P3	work mode code	0x2A	Please check page-18 below.
Pic_ID	Sequence number of the current picture	0x00 0x00	Picture - 00
CC 33 C3 3C	Frame end		Each command must have the frame head and the frame end.

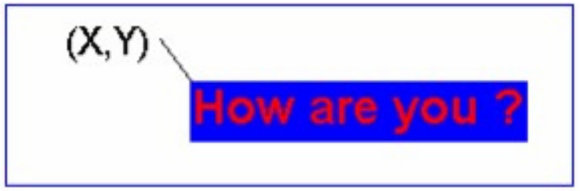
A. Text Command

(1) “0x54”，“0x55”，“0x6E” and “0x6F” Command

Function: The function of “0x54”，“0x55”，“0x6F” and “0x6E” command” is to display 8*16, 16*32,6*12 and 12*24 dot matrix character.

Note: 1. these commands are the default set in the terminal. The character is ASCII character.
2. All of code should be Hexadecimal number
3. The value of the “X” (or “Y”) coordinates should be 2 Bytes Hex number. (For example: (0,1) → 0x00 0x00 0x00 0x01)
4. “String” is the character code which must be **1 Bytes Hexadecimal Unicode code**. For example:” A”→”0x 41”

Command: “**0xAA+0x55+0xX+0xY+ String + 0xCC 0x33 0xC3 0x3C**”
For example: AA 55 00 00 00 01 41 42 43 **CC 33 C3 3C** **(Display ‘ABC’ at (0,1) coordinates position)**



(2) "0x98" Command

Function: The "0x98 command" function is to display various characters.

Note: 1. Make a new font file and create a new Font Lib by yourself. (Please see the document "Creating Font Lib" in the folder-04)
2. The font size and type are decided by the Font File that you created.
3. All of code should be Hexadecimal number.
4. "String" is the character code which must be **2 Byes Hexadecimal Unicode code**. **For example:" A"→"0x00 0x41"**

Command: "0xAA+0x98+0xX+0xY+Lib_ID+C_Mode+C_dots+Color+Bcolor+String+0xCC 0x33 0xC3 0x3C"

For example: AA 98 00 00 00 64 02 C5 03 F8 00 00 00 00 41 00 42 00 43 CC 33 C3 3C (Display the red 'ABC' at (0,100) coordinates position)

Explain:

	Introduction	Example	Note
AA	Frame head		Every command must have the frame head and the frame end
0x98	Command code		
X+Y	Coordinates	(0,100) → 00 00 00 64 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
Lib_ID	Font Lib serial number	The Arial_12_24.dzk file is saved into the second location of the font Lib. So the Lib_ID is "0x02".	
C_Mode	Foreground and Background Color Display mode	If to display both foreground and background colors, the C_Mode is "0xC5"; If to display the foreground colors and not to display background colors, the C_Mode is "0x85";	If you like to cover the last character, please use "0xC5". If you don't like to cover the last character, please use "0x85".
C_dots	Character Dot Matrix size	Take the GREEK3232.DZK file for example. Its Dot Matrix font size is "12*24", so C_dots is "0x03".	Refer picture 02 below
Color+Bcolor	Foreground and Background colors	Red: F8 00 Black: 00 00	You can use this option to check the HEX number of color. (See picture 03)
String	Character	"Hi" → "0x00 0x48 0x00 0x69" 2 Bytes HEX	The character code should be 2 Bytes Unicode Hexadecimal code.
CC 33 C3 3C	Frame end		Every command must have the frame head and the frame end.

Picture 02

C_Dots	Dot Matrix
0x00	8*8
0x01	6*12
0x02	8*16
0x03	12*24
0x04	16*32
0x05	20*40
0x06	24*48
0x07	28*56
0x08	32*64
0x09	12*12
0x0A	16*16
0x0B	24*24
0x0C	32*32
0x0D	40*40
0x0E	48*48
0x0F	56*56
0x10	64*64
0x11	40*80
0x12	48*96
0x13	56*112
0x14	64*128
0x15	80*80
0x16	96*96
0x17	112*112
0x18	128*128
0x19	6*8
0x1A	8*10
0x1B	8*12
0x1C	100*200
0x1D	200*200

B. Picture Command

(1) “0x70” Command

Function: The function of “0x70 command” is to switch pictures.

Note: 1. the first process is to store the picture into the terminal by “Assistant software”.

a. Communicate
b. Choose Pictures
c. Adjust picture sequence
d. Press the button ”Pic Download” .
e. Then picture starts to be delivered to the flash memory, and it will jump up a dialog box as finishing.

2. The image resolution should be same with the terminal screen resolution.

Command: “**0xAA+0x70+ Picture sequence number + 0xCC 0x33 0xC3 0x3C**”

For example: AA 70 00 00 CC 33 C3 3C (Display Picture-00)

2 Bytes Hex

(2) "0x71" Command

Function: The function of "0x71 command" is to display a part of picture-X on the present interface.

Note: 1. the first process is to store the picture into the terminal by "Assistant software".
2. The image resolution should be same with the terminal screen resolution.

Command: "0xAA+0x71+ Picture sequence number +0xX_1+0xY_1+0xX_2+0xY_2 +0xX+0xY+0xCC 0x33 0xC3 0x3C"

For example: AA 71 00 01 00 00 00 00 00 64 00 64 00 32 00 32 CC 33 C3 3C

2 Bytes Hex

(Display the area of picture-01 between (0,0) and (100,100) at (50,50) coordinates position of present interface.)

Explain:

	Introduction	Example	Note
AA	Frame head		Every command must have the frame head and the frame end
0x71	Command code		
Picture serial number	Picture that you want to cut out.	00 01 2 Bytes HEX	Display a part of picture-01.
0xX_1+0xY_1	Top Left Coordinates	(0,0) → 00 00 00 00 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
0xX_2+0xY_2	Lower Right Coordinates	(100,100) → 00 64 00 64 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
0xX+0xY	Display position of present interface	(50,50) → 00 32 00 32 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
0xCC 0x33 0xC3 0x3C"	Frame end		



(3) “0x9C” & “0x9D” Command

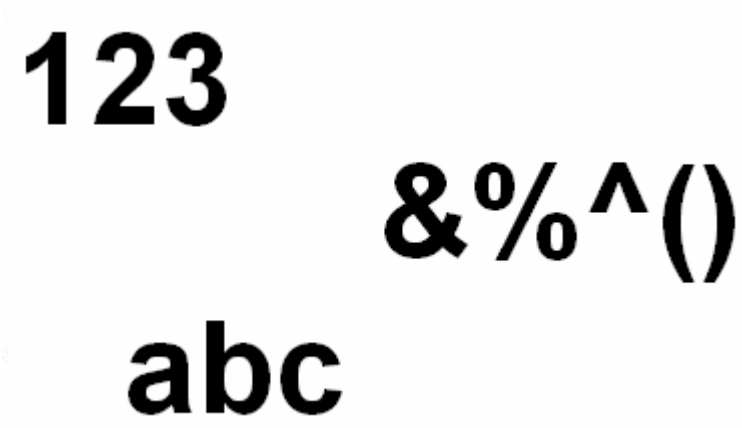
Function: The function of “0x9C” & “0x9D”command” is to display the figure of picture-X on the present interface.

Note: 1. the first process is to store the picture into the terminal by “Assistant software”.
2. The image resolutions should be same with the terminal screen resolution.
3. 0x9C” & “0x9D”command” only displays the single character and the figure on the present interface.
The background color of picture-X is not displayed, and the background color should be single color.
4. Difference between “0x9D” & “0x9C”command”
When there is any other figure on the present interface,
“0x9Ccommand”: Recover the present interface, and then display a part of picture-X.
“0x9D”command”: Display a part of picture-X directly.

Command: “**0xAA+0x9C+ Picture sequence number +0xX_1+0xY_1+0xX_2+0xY_2 +0xX+0xY+0xCC 0x33 0xC3 0x3C**”

For example: AA 9C 00 01 00 00 00 00 00 00 00 64 00 32 00 32 CC 33 C3 3C
(Display the character of picture-01 between (0,0) and (0,100) at (50,50) coordinates position)

Picture-05 ↓ :



(4) “0x60” , “0x61”, “0x62” and “0x63” Command

Function: These commands function is to subject an area to translation.

Note: 1. These command is to display the caption or the figure in a cycle.

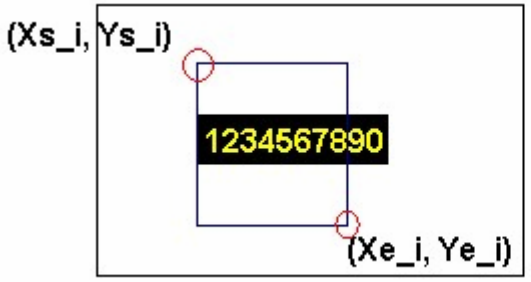
2. “Width” is Translation Width which is from **0x01 to 0x0F**

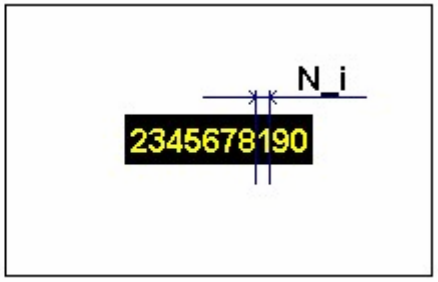
Command: “0xAA+0x60+ 0xX_1+0xY_1+0xX_2+0xY_2 + width +0xCC 0x33 0xC3 0x3C”

For example: AA 60 00 00 00 00 00 64 00 64 0F CC 33 C3 3C

Explain:

		Introduction	Example	Note
AA		Frame head		Every command must have the frame head and the frame tail
Command	0x60	Translation from right to left, Move the leftmost area to the rightmost area		
	0x61	Translation from left to right, Move the rightmost area to the leftmost area		
	0x62	Translation from right to left, the rightmost area is covered by the background color.		
	0x63	Translation from left to right, the leftmost area is covered by the background color.		
0xX_1+0xY_1		Top Left Coordinates	(0,0) → 00 00 00 00 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
0xX_2+0xY_2		Lower Right Coordinates	(100,100) → 00 64 00 64 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
width		Translation Width	0x0F	Move 0x0F width
0xCC 0x33 0xC3 0x3C”		Frame end		





C. Graph Command

(1) "0x51" Command (Point)

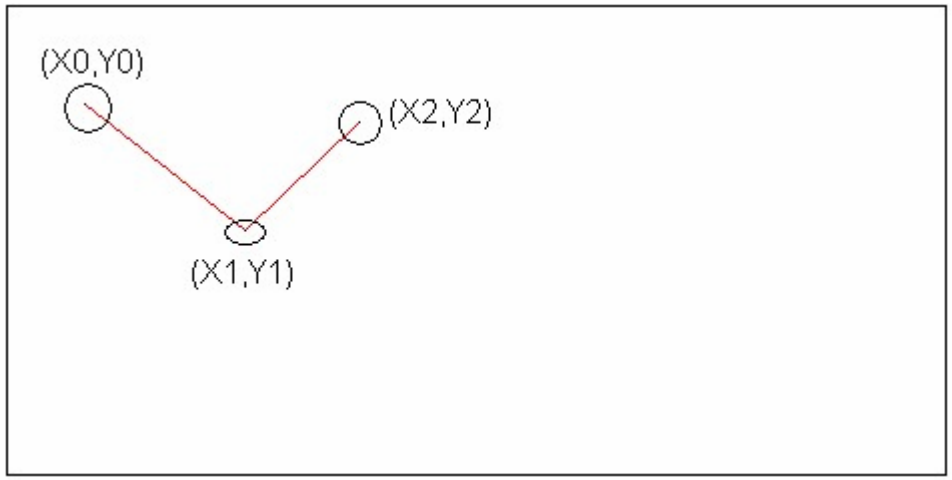
Function: The function of "0x51 command" is to display one point or many points.
Note: 1." **$0xX_1+0xY_1$**" is the coordinates value of the point.

Command: "**0xAA+0x51+ $0xX_1+0xY_1+0xX_2+0xY_2$ +0xCC 0x33 0xC3 0x3C**"
For example: AA 51 00 64 00 64 00 32 00 32 CC 33 C3 3C (Display the point of (100,100) coordinates position and (50,50) coordinates position.)

(2) "0x56" Command (Line)

Function: The function of "0x56 command" is to display one line or many lines.
Note: 1." **$0xX_1+0xY_1$**" is the coordinates value of the point.

Command: "**0xAA+0x56+ $0xX_0+0xY_0+0xX_1+0xY_1$ + $0xX_2+0xY_2$ +$0xX_0+0xY_0$+0xCC 0x33 0xC3 0x3C**"
For example: AA 56 00 14 00 14 00 50 00 5A 00 32 00 AA 00 14 00 14 CC 33 C3 3C
(Connect (20,20)point, (80,90) point and (50,170)point, so that the image is a triangle)



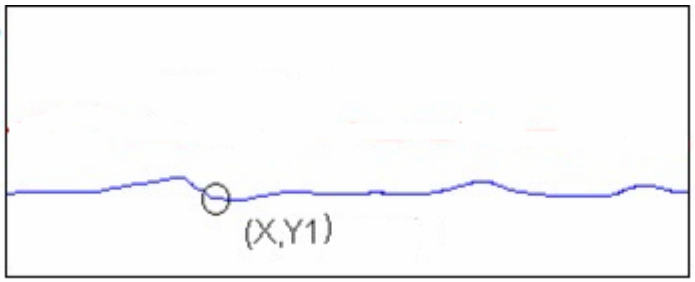
(3) "0x76" Command (Curve)

Function: The function of "0x76 command" is to display a curve.
Note:

Command: "0xAA+0x76+ 0xX+X_add+0xY$_1$+0xY$_2$ +0xY$_3$ + ~~+0xY$_i$+0xCC 0x33 0xC3 0x3C"
For example: AA 76 00 00 01 00 14 00 50 00 5A 00 20 00 AA 00 14 CC 33 C3 3C
(Connect (0,20), (1,80) ,(2,90),(3,32)(4,170)and (5,20)point, so that the image is a curve.)

Explain:

	Introduction	Example	Note
AA	Frame head		Every command must have the frame head and the frame tail
0x76	Command code		
0xX	X-axis coordinates value of starting point	00 00	The value of the X coordinates should be 2 Bytes Hex number.
X_add	Increment of X-axis coordinates (Any increment number is optional)	01	The X-axis coordinates value of each point is 1 more than last point's.
0xY$_1$+0xY$_2$ +0xY$_3$ + ~~+0xY$_i$	Y-axis coordinates value of each point	100 → 00 64 HEX Y	The value of the Y coordinates should be 2 Bytes Hex number.
0xCC 0x33 0xC3 0x3C"	Frame end		



(4) "0x57" Command(Circle)

Function: The function of "0x57 command" is to display one circle or many circles.

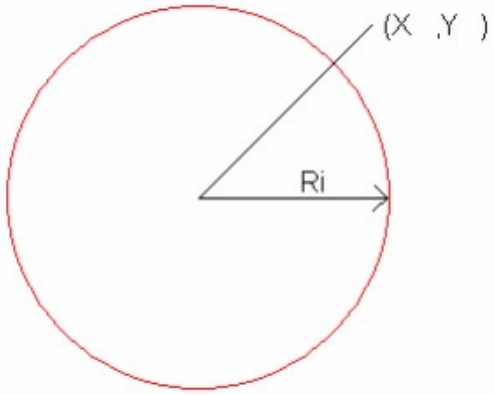
Note:

Command: "0xAA+0x57+Type+0xX+0xY+ Radius +0xCC 0x33 0xC3 0x3C"

For example: AA 57 01 00 14 00 50 14 CC 33 C3 3C (Display a circular ring with 20 radius at (20,80) coordinates position)

Explain:

	Introduction		Example	Note
AA	Frame head			Every command must have the frame head and the frame tail
0x57	Command code			
Type	00	Display circular ring with background color	01	Display circular ring with background color
	01	Display circular ring with foreground color		
	02	Display circular area with background color		
	03	Display circular area with foreground color		
0xX+0xY	Center of a circle		(20,80) → 00 14 00 50 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
Radius	Radius value		"20" → "0x14" HEX	The value of the Radius should be 1 Bytes Hex number
0xCC 0x33 0xC3 0x3C"	Frame end			



(5) "0x59"Command (Rectangle)

Function: The function of "0x57 command" is to display one rectangle or many rectangles.
Note:

Command: "0xAA+0x59+ 0xX_1+0xY_1+0xX_2+0xY_2 +0xCC 0x33 0xC3 0x3C"
For example: AA 59 00 14 00 14 00 50 00 5A CC 33 C3 3C (Display a rectangle, Top left point is (20,20) and lower right point is (80,90))

D. Color command

(1) "0x40" Command (foreground and background color)

Function: The function of "0x40 command" is to set the foreground and background color.
Note: 1.You can check the Hex code of the color by the function of "Assistant Software".
2. The default foreground color is white (0xFFFF), and the default background color is blue(0x001F).

Command: "0xAA+0x40+ Foreground color + Background color +0xCC 0x33 0xC3 0x3C"
For example: AA 40 F8 00 00 1F CC 33 C3 3C (The foreground color is red (0xF800), and the background is blue (0x001F).)



(2) "0x52" Command

Function: The function of "0x52 command" is to clear the screen.

Note:

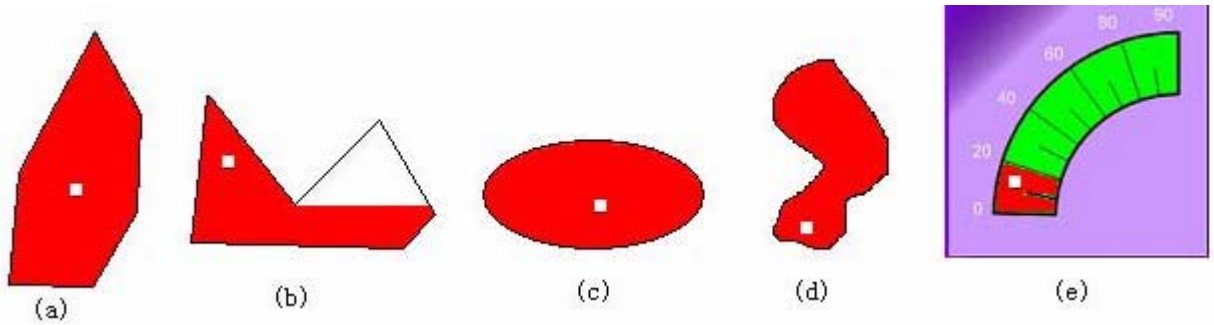
Command: "**0xAA+0x52+0xCC 0x33 0xC3 0x3C**"
For example: **AA 52 CC 33 C3 3C** **(Clear the screen)**

(3) "0x64" Command

Function: The function of "0x64 command" is to fill the irregular area with different color.
Note: 1. the color of this irregular area should be same with the color of (X, Y) point, otherwise it only fill the area which color is the same with the color of point.
2. This command can't change the foreground and background color.

Command: "**0xAA+0x64+ 0xX+0xY+ Color +0xCC 0x33 0xC3 0x3C**"

For example: AA 64 00 14 00 14 F8 00 CC 33 C3 3C **(Fill this area with red)**



(4) "0x9B" & "0xE7"Command (Clock)

Function: The function of "0x9B command" is to close and open clock function.
The function of "0xE7 command" is to adjust clock.

a. **Close Clock.**
Command: "0xAA+0x9B + 0x00 +0xCC 0x33 0xC3 0x3C"
For example: AA 9B 00 CC 33 C3 3C

b. **Open Clock**
Command: "0xAA+0x9B + 0xFF + RTC_Mode + Text_Mode + Color + 0xX+0xY +0xCC 0x33 0xC3 0x3C"
For example: AA 9B FF 00 03 F8 00 00 00 00 64 CC 33 C3 3C (Fill this area with red)

Explain:

	Introduction		Example	Note
AA	Frame head			Every command must have the frame head and the frame end
0x9B 0xFF	Command code			
RTC_Mode	0x00	HH:MM:SS	0x00 mode: 15:30:00	
	0x01	20YY:MM:DD HH:MM:SS		
Text_Mode	0x00	6*12	0x03 mode: 16*32 dot matrix font	
	0x01	8*16		
	0x02	12*24		
	0x03	16*32		
	0x04	24*48		
	0x05	32*64		
Color	Setting color		Red: F8 00	
X+Y	Coordinates		(0,100) → 00 00 00 64 HEX X Y	The value of the X (or Y) coordinates should be 2 Bytes Hex number.
CC 33 C3 3C	Frame end			

c. **Adjust Clock**
Command: "0xAA+ 0xE7+0x55+0xAA+0x5A+0xA5 + HH+MM+SS +0xCC 0x33 0xC3 0x3C"
For example: AA E7 55 AA 5A A5 0F 1E 00 CC 33 C3 3C (Display "15:30:00")
Note: All the number of time should be "Hex number".

d. **Read off current Clock**
Send Command: "0xAA+ 0x9B+0x5A +0xCC 0x33 0xC3 0x3C"
Receive Command: "0xAA+ 0x9B+0x5A +YY+MM+DD+WW+HH+MM+SS +0xCC 0x33 0xC3 0x3C"

For example: Receive AA 9B 5A 08 0C 19 04 09 3A 00 CC 33 C3 3C (Mean 09:58:00, Thursday, Dec. 25th, 2008.)
Note: All the number of time should be "Hex number".

(5) "0x5E" & "0x5F"Command (Backlight)

Function: The function of "0x5E command" is to close the backlight.
The function of "0x5F command" is to open and adjust the backlight.

a. **Close Backlight.**
Command: "0xAA+0x5E +0xCC 0x33 0xC3 0x3C"
For example: AA 5E CC 33 C3 3C

b. **Open Backlight**
Command: "0xAA+0x5F +0xCC 0x33 0xC3 0x3C"
For example: AA 5F CC 33 C3 3C

c. **Adjust Backlight**
Command: "0xAA+ 0x5F + PWM_T +0xCC 0x33 0xC3 0x3C"
For example: AA 5F 36 CC 33 C3 3C
Note: 1. "PWM_T" mean the setting value of backlight, which range is from 0x00 to 0x63.
"0x00" mean closing backlight, and "0x63" mean the backlight is the biggest.
2. The regulation will not be saved, the terminal defaults the biggest backlight after turning on.

E. Touch-Screen Command

(1) “0x72” and “0x73”Command (Feedback Value from the terminal)

Function: If you **don’t** use the touch-control file,
The terminal outputs feedback to your MCU by “0x73 command” when your finger presses the touch-screen. (Once every 100ms or only once)
The terminal outputs feedback to your MCU by “0x72 command” when your finger releases. (Only once.)

Note: 1.The feedback value is the coordinates value of your touch position.
2. The terminal outputs feedback value to your MCU automatically.
3. Your MCU can perform the related program in accordance with this “coordinates value”.
(For example: There is a button in this area that you touch.)

Feedback Command: “0xAA+0x73+0xX+0xY+0xCC 0x33 0xC3 0x3C”
“0xAA+0x72+0xX+0xY+0xCC 0x33 0xC3 0x3C”

For example: AA 73 00 14 00 50 CC 33 C3 3C (The terminal output feedback of (20,80) to your MCU when your finger presses the screen.)
AA 72 00 14 00 50 CC 33 C3 3C (The terminal output feedback of (20,80) to your MCU when your finger releases.)



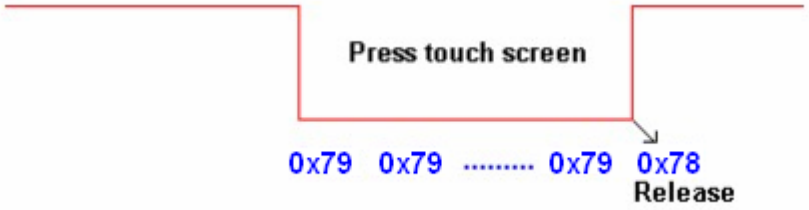
(2) “0x79” Command and “0x78” Command (Feedback from the terminal)

Function: If you use the touch control file,
The terminal outputs feedback to your MCU by “0x79 command” when your finger presses the touch-screen. (Once every 100ms or only once)
The terminal outputs feedback to your MCU by “0x78 command” when your finger releases. (Only once.)

Note: 1. Make a touch control files by “Touch-Control software” firstly.
2. Each button on the screen has a feedback value which is set through “Touch-Control software” by yourself.
3. Your MCU can perform the relative program in accordance with the feedback value.
4. You can use “0xE0 command” or “Assistant software” to change the working mode configuration below.
5. The feedback value is 2 byte Hex number.
Please see the document “Create Touch Control file” in the “03-Touch Control manual” folder.

Command: “**0xAA+0x79+Feedback Value+0xCC 0x33 0xC3 0x3C**”
“**0xAA+0x78+ Feedback Value+0xCC 0x33 0xC3 0x3C**”

For example: **AA 79 00 01 CC 33 C3 3C** **(when your finger presses the touch-screen, the feedback value is 0x01.)**
AA 78 00 01 CC 33 C3 3C **(When your finger releases, the feedback value is 0x01.)**



(3) “0xE0” Command (working mode configuration)

Function: The “0x E0 command” function is to change the working mode configuration.

Note: 1. If you don’t use the touch control file, please don’t forget to set “Para1” to 0x00.
2. Please don’t often change the working mode configuration.

Command: “0xAA+0xE0 +Bode_Set+Para1 +0xCC 0x33 0xC3 0x3C”
For example: AA E0 07 2A CC 33 C3 3C (You may use “79&78 Command”)

Explain:

	Introduction	Example	Note
AA	Frame head		Every command must have the frame head and the frame end
0xE0	Command code		
Bode_Set	Choose the Baud Rate	For example: “0x07” code mean 115200 bps.	Refer to Picture-06
Para1	Work Mode	If to use “78 Command”, the Para1 is 0x2A. If you don’t use “78 Command”, the Para1 is 0x00. “0 0 1 0 0 0 1 0” → “0x2A” Binary code HEX code	Refer to picture-07.
CC 33 C3 3C	Frame end		

Note: About “Para1”, you can choose any work mode in accordance picture-07.
For example: You want to use “78”command, So the “Para1” is binary code “0 0 1 0 0 0 0 0”.(HEX code is 0x20)
You don’t want the buzzer tweet, So the “Para1” is binary code “0 0 0 0 1 0 0 0”.(HEX code is 0x08)
As you only touch the button, the buzzer tweet, so the “Para1” is binary code “0 0 1 0 1 0 1 0”.(HEX code is 0x2A)

Picture-06

Bode_Set	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07
Baud-rate	1200	2400	4800	9600	19200	28400	57600	115200

Picture-07

<Para1>: to set touch panel and keyboard work mode, described as table blow.

Para1	Bit description
.7	0= to upload command 0x72 when release touch panel, it must be 0 when you use touch config file. 1= not to upload command 0x72 when release touch panel.
.6	0= to upload command 0x73 or 0x79 every 100ms until release touch panel. 1= to upload command 0x73 or 0x79 only once when click touch panel.
.5	0= Not use the touch config file to control terminal 1= use the touch config file to control terminal
.4	0
.3	0= buzzer rings when click the screen or the keyboard 1= buzzer does not ring when click the screen or the keyboard
.2	0
.1	0= buzzer rings all the time when you use touch control file. 1= buzzer only rings when you touch the button that you set in the touch control file. (Note: “Para1-3”should be “1”.)
.0	0= not to upload command 0x79 1= to upload command 0x79 (Note: Please select it as you use the 0x79 command.)